

LLM Steering Vectors for Physics

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Abstract

This paper tests whether activation steering improves a small Qwen language model on MMLU-Pro Physics questions. Correct and incorrect unsteered validation generations are mined, paired, and used to train layer-specific steering vectors from final-token activations. Each steering vector is computed as a mean activation difference, then applied by adding it to a selected model layer with a chosen scaling factor. Steered held-out test accuracy is compared with an unsteered baseline.

1 Mathematical Formulation

What follows is mathematization of steering vectors [1].

Let $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Let an alphabet be a finite nonempty set of symbols. Let A^* denote the Kleene closure of an alphabet A . Let $\Delta(E)$ be the set of probability distributions over a finite or countable set E . Let Θ be the model-parameter space.

Let Σ_{text} be a text alphabet and Σ_{tok} be a token alphabet. Let $\mathcal{S} = \Sigma_{\text{text}}^*$ be the set of finite text strings and $\mathcal{X} = \Sigma_{\text{tok}}^*$ be the set of finite token strings. Let $\text{tok} : \mathcal{S} \rightarrow \mathcal{X}$ be the tokenizer. Let $\text{detok} : \mathcal{X} \rightarrow \mathcal{S}$ be the detokenizer, with $\text{detok}(\text{tok}(s)) = s$ for all $s \in \mathcal{S}$, so detok is the inverse of tok on $\text{tok}(\mathcal{S})$. Let $\mathcal{V} = \{\eta_1, \dots, \eta_R\}$ be a finite answer-label alphabet of size $R \in \mathbb{N}$.

Let $V \in \mathbb{N}$ be the number of sampled completions per validation prompt, $T \in \mathbb{R}_{>0}$ be temperature, and $q \in (0, 1]$ be the nucleus sampling threshold. Let $\theta \in \Theta$, and let $P_\theta : \mathcal{X} \rightarrow \Delta(\Sigma_{\text{tok}})$ be the next-token conditional distribution map of the unsteered autoregressive model. Let the token-completion and text-completion conditional distribution maps be

$$M_\theta^{\text{tok}} : \mathcal{X} \times \mathbb{R}_{>0} \times (0, 1] \rightarrow \Delta(\mathcal{X}), \quad M_\theta : \mathcal{S} \times \mathbb{R}_{>0} \times (0, 1] \rightarrow \Delta(\mathcal{S}).$$

For $z \in \mathcal{X}$, let $M_\theta^{\text{tok}}(\cdot \mid z; T, q)$ be the induced distribution on token-string completions in $\mathcal{X}$. For $s \in \mathcal{S}$ and $B \subseteq \mathcal{S}$, define the decoded text-completion distribution by

$$M_\theta(B \mid s; T, q) = M_\theta^{\text{tok}}(\{x \in \mathcal{X} : \text{detok}(x) \in B\} \mid \text{tok}(s); T, q).$$

Let $u \in \mathbb{N}$ be the number of validation rows and $I = \{1, \dots, u\}$ be the validation index set. Let $i \in I$ index validation rows, $p_i \in \mathcal{S}$ be validation

prompt i , and $a_i \in \mathcal{Y}$ be the gold validation answer. Let $K = \{1, \dots, V\}$ be the sampled-completion index set, and let k index sampled completions in K . Let $g_{ik} \in \mathcal{S}$ be sampled by

$$g_{ik} \sim M_\theta(\cdot \mid p_i; T, q).$$

Let $\hat{a} : \mathcal{S} \rightarrow \mathcal{Y} \cup \{\emptyset\}$ be answer extraction, $\emptyset$ be no extracted answer, and $\parallel : \mathcal{S} \times \mathcal{S} \rightarrow \mathcal{S}$ be string concatenation. Let $P, N \subseteq \mathcal{S}$ be correct text and incorrect parseable text:

$$\begin{aligned} P &= \{p_i \parallel g_{ik} : (i, k) \in I \times K, \\ &\quad \hat{a}(g_{ik}) = a_i\}, \\ N &= \{p_i \parallel g_{ik} : (i, k) \in I \times K, \\ &\quad \hat{a}(g_{ik}) \neq a_i, \hat{a}(g_{ik}) \neq \emptyset\}. \end{aligned}$$

Let $|\cdot|$ be cardinality. Let $n = \min(|P|, |N|) \in \mathbb{N}$ be the pair count. Let $P^\pi = (p_1^\pi, \dots, p_{|P|}^\pi) \in \mathcal{S}^{|P|}$ and $N^\rho = (n_1^\rho, \dots, n_{|N|}^\rho) \in \mathcal{S}^{|N|}$ be random shuffles of P and N . Let $j \in \{1, \dots, n\}$ index pairs and $D \in (\mathcal{S} \times \mathcal{S})^n$ be the paired contrast sequence:

$$D = ((p_1^\pi, n_1^\rho), \dots, (p_n^\pi, n_n^\rho)).$$

Let $\mathcal{L} \subset \mathbb{N}_0$ be the steered decoder layers. Let $\ell \in \mathcal{L}$ be a decoder layer and $t \in \mathbb{N}_0$ be a token position. Let $d \in \mathbb{N}$ be the activation dimension, $x \in \mathcal{S}$ be text, $\tau : \mathcal{S} \rightarrow \mathbb{N}_0$ be the read-token map on $\text{tok}(x)$, $h_{\ell,t}(x) \in \mathbb{R}^d$ be the decoder-block activation on $\text{tok}(x)$, and $v_\ell \in \mathbb{R}^d$ be the steering vector:

$$\begin{aligned} \tau(x) &= \text{last non-padding position of } \text{tok}(x), \\ v_\ell &= \frac{1}{n} \sum_{j=1}^n \left(h_{\ell, \tau(p_j^\pi)}(p_j^\pi) - h_{\ell, \tau(n_j^\rho)}(n_j^\rho) \right). \end{aligned}$$

Let $\mathcal{A} \subset \mathbb{R}_{>0}$ be the set of scaling multipliers, and let $\alpha \in \mathcal{A}$ be a multiplier. Let b be an unsteered baseline symbol with $b \notin \mathcal{L} \times \mathcal{A}$. Let $\mathcal{C} = \{b\} \cup (\mathcal{L} \times \mathcal{A})$ be the condition set and $c \in \mathcal{C}$ be a condition. Let $\lambda \in \mathbb{N}_0$ index decoder layers and $h_{\lambda,t}^{(c)}(x) \in \mathbb{R}^d$ be the condition-specific activation:

$$\begin{aligned} h_{\lambda,t}^{(b)}(x) &= h_{\lambda,t}(x), \\ h_{\lambda,t}^{(\ell, \alpha)}(x) &= \begin{cases} h_{\ell,t}(x) + \alpha v_\ell, & \lambda = \ell, \\ h_{\lambda,t}(x), & \lambda \neq \ell. \end{cases} \end{aligned}$$

Let $M_\theta^{(c)}$ be M_θ run with activations $h_{\lambda,t}^{(c)}$, and write $M_\theta^{(c)}(s)$ for its generated text completion on $s \in \mathcal{S}$. Let $m \in \mathbb{N}$ be the number of test rows. Let $r \in \{1, \dots, m\}$ index test rows, $s_r \in \mathcal{S}$ be the r -th test prompt, $a_r \in \mathcal{Y}$ be the r -th gold test answer, and $y_r^{(c)} \in \mathcal{S}$ be the evaluation completion:

$$y_r^{(c)} = M_\theta^{(c)}(s_r).$$

Let $\mathbf{1}[E] \in \{0, 1\}$ be 1 if event E is true and 0 otherwise, $\text{Acc} : \mathcal{C} \rightarrow [0, 1]$ be accuracy, and $\Delta_{\ell, \alpha} \in [-1, 1]$ be accuracy delta versus baseline:

$$\text{Acc}(c) = \frac{1}{m} \sum_{r=1}^m \mathbf{1}[\hat{a}(y_r^{(c)}) = a_r], \quad \Delta_{\ell, \alpha} = \text{Acc}(\ell, \alpha) - \text{Acc}(b).$$

Paper’s Codebase Uses

Paper’s codebase uses $M_\theta = \text{Qwen3.5-0.8B}$, a Qwen3.5 autoregressive transformer. It sets

$$V = 4, \quad T = 0.8, \quad q = 0.95,$$

$$\mathcal{L} = \{6, 12, 18\}, \quad \mathcal{A} = \{0.5, 1.0, 1.5, 2.0\}.$$

References

- [1] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. *Steering Language Models With Activation Engineering*. arXiv:2308.10248, 2023. <https://arxiv.org/abs/2308.10248>.